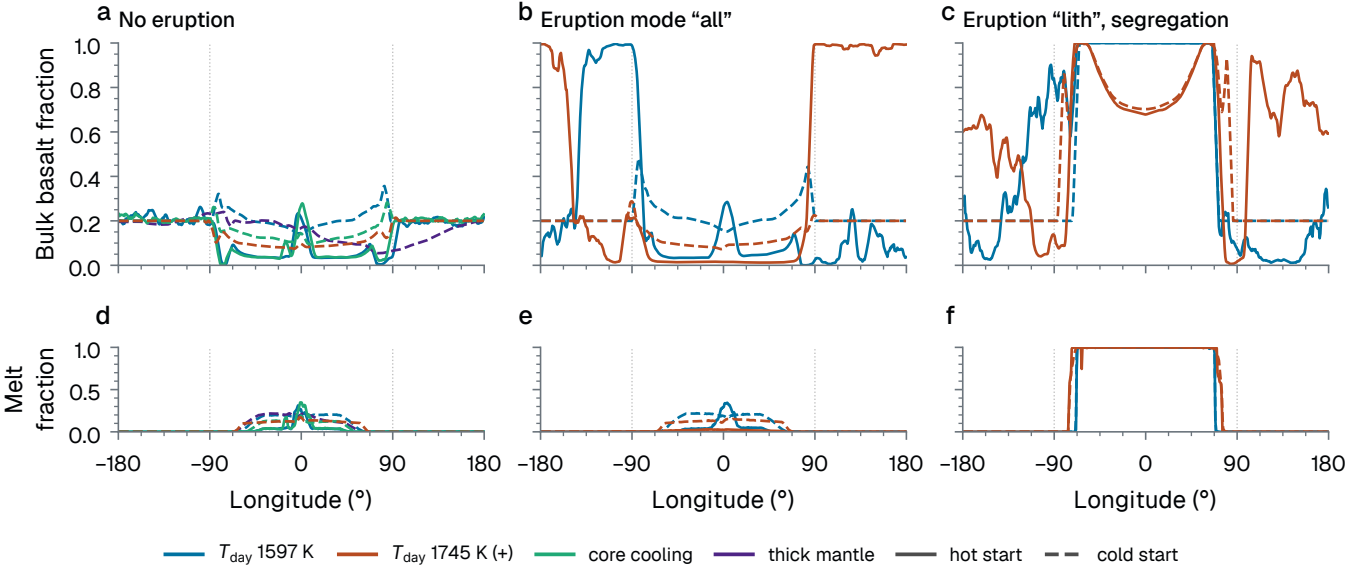

a No eruption
b Eruption mode “all”
c Eruption “lith”, segregation
Bulk basalt fraction
1.0
0.8
0.6
0.4
0.2
0.0
d
e
f
Melt fraction
1.0
0.5
0.0
−180
−90
0
90
180
Longitude (°)
T_{day} 1597 K
T_{day} 1745 K (+)
core cooling
thick mantle
hot start
cold start